

Essential ideas

9.1

Redox (reduction–oxidation) reactions play a key role in many chemical and biochemical processes.

9.2
and
19.1

Voltaic cells convert chemical energy to electrical energy and electrolytic cells convert electrical energy to chemical energy. Energy conversions between electrical and chemical energy lie at the core of electrochemical cells.

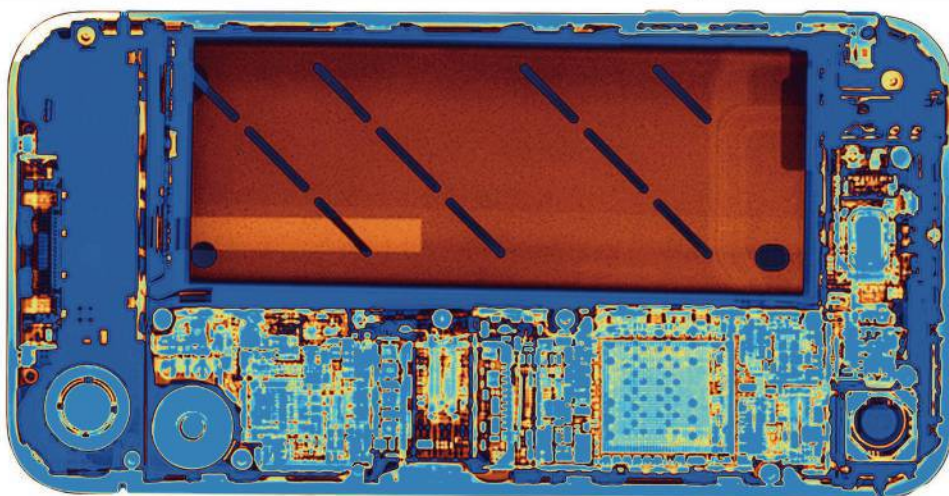
Oxygen makes up only about 20% by volume of the air, yet it is the essential component for so many reactions. Without it fuels would not burn, iron would not rust, and we would be unable to obtain energy from our food molecules through respiration. Indeed, animal life on the planet did not evolve until a certain concentration of oxygen had built up in the atmosphere over 600 million years ago. The term **oxidation** has been in use for a long time to describe these and other reactions where oxygen is involved in chemical change. Oxidation, though, is only half of the story, as it is always accompanied by the opposite process **reduction**, which was originally thought of in terms of loss of oxygen.

Later, however, the terms widened to include a much broader range of reactions. We now define these two processes, oxidation and reduction, as occurring whenever electrons are transferred from one reactant to another – and many of these reactions do not use oxygen. For example, photosynthesis, the process by which plants store chemical energy from light energy, involves oxidation and reduction reactions, although oxygen itself is not a reactant.

Transferring electrons from one substance to another leads to a flow of electrons, in other words an electric current. So, chemical reactions can be used to generate electricity – a simple **voltaic cell** or **battery** works in this way. By reversing the process and using an electric current to drive reactions of oxidation and reduction, stable compounds can be broken down into their elements. This is the process known as **electrolysis**. These applications of oxidation and reduction, collectively known as **electrochemical cells**, have truly revolutionized our world. For example, it would be hard to imagine life without the many battery-powered mobile devices we use every day, or without metals such as aluminium that are only available from electrolysis. An understanding of oxidation and reduction is therefore at the heart of the study of a large branch of chemistry both in the laboratory and beyond.

The vigorous reaction here shows the redox change that occurs when magnesium metal and copper oxide are heated together, forming magnesium oxide and copper. $\text{Mg(s)} + \text{CuO(s)} \rightarrow \text{MgO(s)} + \text{Cu(s)}$. Redox reactions occur when electrons are transferred between species, and are the basis of electrochemistry.

Coloured X-ray of a smart phone, showing its rechargeable battery. Mobile devices such as this are powered by electrical energy generated from the reactions of oxidation and reduction occurring within the battery.



9.1 Oxidation and reduction

Understandings:

- Oxidation and reduction can be considered in terms of oxygen gain/hydrogen loss, electron transfer, or change in oxidation number.
- An oxidizing agent is reduced and a reducing agent is oxidized.

Guidance

Oxidation states should be represented with the sign given before the number, e.g. +2 not 2+.

- Variable oxidation numbers exist for transition metals and for most main-group non-metals.

Guidance

The oxidation state of hydrogen in metal hydrides (−1) and oxygen in peroxides (−1) should be covered.

- The activity series ranks metals according to the ease with which they undergo oxidation.

Guidance

A simple activity series is given in the IB data booklet in section 25.

- The Winkler method can be used to measure biochemical oxygen demand (BOD), used as a measure of the degree of pollution in a water sample.

Applications and skills:

- Deduction of the oxidation state of an atom in an ion or a compound.
- Deduction of the name of a transition metal compound from a given formula, applying oxidation numbers represented by Roman numerals.

Guidance

Oxidation number and oxidation state are often used interchangeably, though IUPAC does formally distinguish between the two terms. Oxidation numbers are represented by Roman numerals according to IUPAC.

- Identification of the species oxidized and reduced and the oxidizing and reducing agents, in redox reactions.
- Deduction of redox reactions using half-equations in acidic or neutral solutions.
- Deduction of the feasibility of a redox reaction from the activity series or reaction data.
- Solution of a range of redox titration problems.
- Application of the Winkler method to calculate BOD.

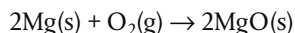
Introduction to oxidation and reduction

Early attempts to define oxidation and reduction were based on observations of the gain and loss of oxygen and hydrogen during chemical change. They can be summarized as follows.

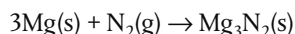
Oxidation	Reduction
gain of oxygen	loss of oxygen
loss of hydrogen	gain of hydrogen

These definitions are still useful in some cases, particularly in organic chemistry, but they have largely been replaced by a more inclusive approach. It is now recognized that oxidation and reduction occur during chemical change whenever there is a shift in electron density from one atom to another. In other words, the focus has become the transfer of electrons, whether complete or partial, whether oxygen and hydrogen are involved or not. Some examples will help to explain this.

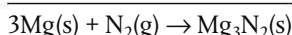
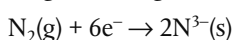
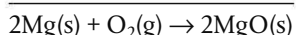
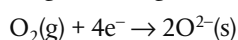
When magnesium is burned in air, it gives a bright white flame with the production of a white powder, magnesium oxide:



The fact that magnesium gains oxygen in this reaction makes it easy to see why we say it is an **oxidation** reaction and that magnesium has been oxidized. However, during the same reaction a small amount of the magnesium combines with the nitrogen of the air too, forming magnesium nitride. It may be less obvious that this is also an oxidation reaction and that again magnesium is oxidized.



What do these two reactions have in common which means they can both be defined in this way? If we divide them into so-called **half-equations**, each showing what happens to one reactant, we can examine what is happening in terms of electrons.



In both reactions Mg is forming Mg^{2+} by losing electrons, while O and N are forming O^{2-} and N^{3-} respectively by gaining electrons. It is this transfer of electrons that now defines oxidation, and its opposite reaction, reduction.

🔥 Oxidation is the loss of electrons, reduction is the gain of electrons. 🍷

So in the reactions above, magnesium is oxidized while oxygen and nitrogen are reduced. Clearly each process is dependent on the other, so oxidation and reduction will always occur together and reactions of this type are known as **redox reactions**.

The free-radical theory of ageing suggests that the physiological changes associated with ageing are the result of oxidative reactions in cells causing damage to membranes and large molecules such as DNA. These changes accumulate with time and may explain the increase in degenerative diseases, such as cancer, with age. The theory suggests that supplying cells with anti-oxidants will help to slow down the damaging oxidative reactions. Anti-oxidants are particularly abundant in fresh fruit and vegetables, as well as in red wine, tea, and cocoa. Although there is strong evidence that anti-oxidant supplementation may help protect against certain diseases, it has not yet been shown to produce a demonstrated increase in the human life span.



▲ Fresh fruits and vegetables are good sources of anti-oxidants, which may help prevent damaging oxidative reactions in cells.

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▲ Magnesium ribbon burning in air. It forms mostly magnesium oxide with small amounts of magnesium nitride.



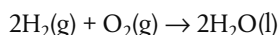
It may help you to remember OILRIG: Oxidation Is Loss (of electrons), Reduction Is Gain (of electrons).



Oxidation is the loss of electrons, reduction is the gain of electrons.

Oxidation numbers enable us to track redox change

In reactions involving ions, such as the examples above, it is easy to identify the electron transfers occurring. But what about reactions where electrons are not transferred but instead are shared in a covalent bond, such as the combination of hydrogen and oxygen to produce water? Can oxidation and reduction be identified here too?



The answer is 'yes', through the introduction of the concept of **oxidation state**. This is a value we assign to each atom in a compound that is a measure of the electron control or possession it has relative to the atom in the pure element. Oxidation state can be defined as the apparent charge that an atom has in a molecule or ion. It is as if we exaggerate the unequal sharing in a covalent bond, based on electronegativity differences, to the point where each atom has complete gain or loss of the electrons shared – in other words, has formed ions. This enables us to keep track of the relative electron density in a compound and how it changes during a reaction. There are two parts to the oxidation state:

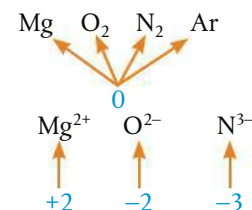
- the sign: '+' means the atom has lost electron control, '-' means it has gained electron control;
- its value: this refers to the number of electrons over which control has changed.

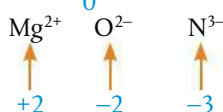
The oxidation state is written with the sign first, followed by the number, e.g. +2 or -3.

Strategy for assigning oxidation states

There are a few simple rules to follow which help in determining the oxidation state of an atom in any species. We will use blue to show oxidation states throughout this text.

- 1 Atoms in the free (uncombined) element have an oxidation state of zero: for example,


- 2 In simple ions, the oxidation state is the same as the charge on the ion: for example,


- 3 The oxidation states of all the atoms in a neutral (uncharged) compound must add up to zero:

for example, in H_2SO_4 the sum of oxidation states = 0.
- 4 The oxidation states of all the atoms in a polyatomic ion must add up to the charge on the ion:

for example, in SO_4^{2-} the sum of oxidation states = -2.
- 5 The usual oxidation state for an element is the same as the charge on its most common ion:

for example, Group 1 elements have oxidation state = +1, H is usually +1, O is usually -2.
- 6 Most main group non-metals, the elements at the bottom of Group 14, and transition elements have oxidation states that vary in different compounds – depending on the conditions and other elements present. So for N, P, S, Sn, Pb, and all transition elements, the oxidation state of the element in a particular species needs to be determined on a case-by-case basis.

The table on page 409 summarizes some useful information about oxidation states.

Note that the charge on an ion X is written with the number first then the charge: e.g. X^{2+} . The oxidation state is written with the charge first then the number: e.g. +2.



What ways of knowing can we use to distinguish between the charge on an ion and the oxidation state?

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Element	Usual oxidation state	Exceptions	Explanation
Li, Na, K	+1		
Mg, Ca	+2		
F	-1		there are no exceptions because F is the most electronegative element
O	-2	peroxides such as H_2O_2 , where it is -1; OF_2 , where it is +2	
H	+1	metal hydrides such as NaH , where it is -1	H is more electronegative than Na and so gains electron control
Cl	-1	when it is combined with O or F	Cl is less electronegative than O and F, and so loses electron control

When working out the oxidation states of atoms in compounds, it is usually best to assign the oxidation state to the atoms that are easy to predict first, then use Rules 3 and 4 above to find the more unpredictable elements by subtraction.

Worked example

Assign oxidation states to all the elements in (a) H_2SO_4 and (b) SO_3^{2-} .

Solution

(a) We can assign H and O as follows:

$$\begin{array}{c} \text{H}_2\text{SO}_4 \\ \uparrow \quad \uparrow \\ +1 \quad -2 \end{array}$$

Note that the oxidation states apply to each atom and that here the sum of all the oxidation states must be zero as H_2SO_4 is electrically neutral.

Therefore, $2(+1) + S + 4(-2) = 0 \Rightarrow S = +6$

(b) Here we start by assigning O:

$$\begin{array}{c} \text{SO}_3^{2-} \\ \uparrow \\ -2 \end{array}$$

Note that here the oxidation states must add up to -2, the charge on the ion.

Therefore, $S + 3(-2) = -2 \Rightarrow S = +4$

The oxidation state of a transition metal in a complex ion can be worked out from the charges on the ligands and the overall charge. Remember from Chapter 3 that ligands are either neutral or negatively charged.

Worked example

Assign oxidation states to the metal ion in (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ and (b) $[\text{CuCl}_4]^{2-}$.

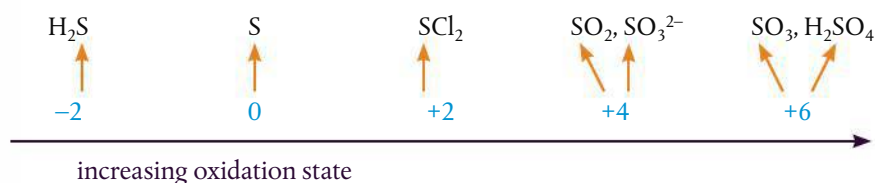
Solution

(a) NH_3 is a neutral ligand, so the charge on the complex is the same as the charge on the metal ion: $\therefore \text{Co} = +3$

(b) Cl has a 1- charge $\therefore (\text{charge on Cu}) + (4 \times 1-) = 2-$
 $\therefore \text{Cu} = +2$

Interpreting oxidation states

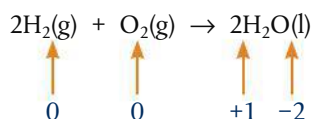
We can see that an element such as sulfur can have a wide range of oxidation states in different compounds.



What is the significance of these different values? Because the oxidation state is a measure of the electron control that an atom has, it follows that the higher the positive number, the more the atom has lost control over electrons, in other words the more oxidized it is. Likewise, the greater the negative number, the more it has gained electron control, so the more reduced it is. Therefore H_2S represents sulfur in its most reduced form (lowest oxidation number), and SO_3 and H_2SO_4 represent sulfur in its most oxidized form (highest oxidation number).

It follows that any change in oxidation states during a reaction is an indication that redox processes are occurring: an increase in oxidation number represents oxidation, and a decrease in oxidation number represents reduction.

So, going back to the reaction between hydrogen and oxygen discussed earlier, we can now clearly follow the redox process:



Hydrogen has been oxidized (oxidation state increased from 0 to +1) and oxygen has been reduced (oxidation state decreased from 0 to -2).

A redox reaction is a chemical reaction in which changes in the oxidation states occur.

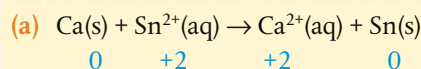
Oxidation occurs when there is an increase in oxidation state of an element, reduction occurs when there is a decrease in oxidation state of an element.

Worked example

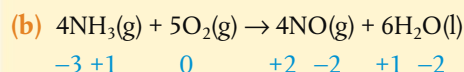
Use oxidation states to deduce which species is oxidized and which is reduced in the following reactions:

- (a) $\text{Ca}(\text{s}) + \text{Sn}^{2+}(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{Sn}(\text{s})$
 (b) $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{l})$

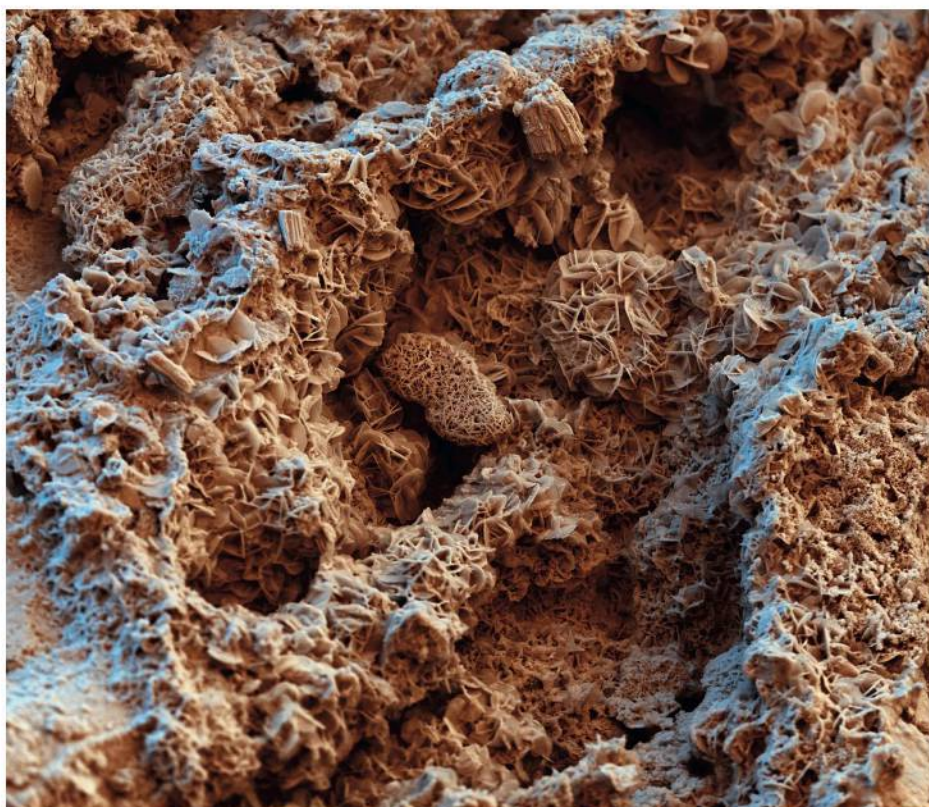
Solution



Ca is oxidized because its oxidation state increases from 0 to +2; Sn^{2+} is reduced because its oxidation state decreases from +2 to 0.



NH_3 is oxidized because the oxidation state of N increases from -3 to +2; O_2 is reduced because its oxidation state decreases from 0 to -2.



Coloured scanning electron micrograph (SEM) of rust on painted sheet metal. Rust is hydrated iron oxide resulting from electrochemical reactions between iron and atmospheric water vapour and oxygen. Rusting spontaneously causes iron to revert to its more stable oxidized state, unless this is prevented by protecting the iron. Rust flakes off the surface of the iron, causing it to be degraded and weakened.

Systematic names of compounds use oxidation numbers

We have seen that elements such as sulfur exhibit different oxidation states in different compounds. In these cases it is useful to include information about the oxidation state in the name. Traditional names used descriptive language that became associated with a particular oxidation state. For example, ferrous and ferric iron oxides referred to FeO and Fe_2O_3 , in which Fe has oxidation states **+2** and **+3** respectively.

The IUPAC system, founded in 1919, introduced a nomenclature using **oxidation numbers** to make the names more recognizable and unambiguous. This system uses a Roman numeral corresponding to the oxidation state which is inserted in brackets after the name of the element. For example copper(I) oxide contains copper in the oxidation state **+1**. Different compounds in which the same elements have different oxidation states are then shown with different oxidation numbers, such as copper(I) oxide and copper(II) oxide.

The table shows some common examples.

Formula of compound	Oxidation state	Name using oxidation number	Formula of compound	Oxidation state	Name using oxidation number
FeO	Fe +2	iron(II) oxide	MnO_4^-	Mn +7	manganate(VII) ion
Fe_2O_3	Fe +3	iron(III) oxide	$\text{K}_2\text{Cr}_2\text{O}_7$	Cr +6	potassium dichromate(VI)
Cu_2O	Cu +1	copper(I) oxide	Cr_2O_3	Cr +3	chromium(III) oxide
CuO	Cu +2	copper(II) oxide	NO	N +2	nitrogen(II) oxide
MnO_2	Mn +4	manganese(IV) oxide	NO_2	N +4	nitrogen(IV) oxide



Oxidation state is shown with a '+' or '-' sign and an Arabic numeral, e.g. **+2**.

Oxidation number is shown by inserting a Roman numeral in brackets after the name or symbol of the element.

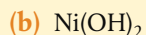
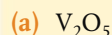
Although we have made the distinction here between oxidation number and oxidation state according to IUPAC guidelines, in reality this is not the main point of this discussion. In fact many sources use oxidation number and oxidation state interchangeably, so it is not too important. What does matter though, is the information represented in these values in terms of the electron density associated with an atom in a compound.

The IUPAC system aims to help chemists communicate more easily in all languages by introducing systematic names into compounds. However, its success in achieving this will be determined by how readily it is adopted. What do you think might prevent chemists using exclusively the 'new names'?



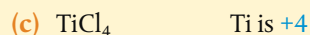
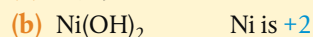
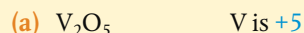
Worked example

Deduce the name of the following compounds using oxidation numbers.

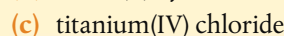
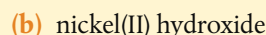
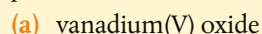


Solution

1 First deduce the oxidation state:



2 Then the corresponding Roman numeral is inserted after the name of the element. There is no space between the name and the number, and the number is placed in brackets.



Although this nomenclature can theoretically be used in the naming of all compounds, it is really only worthwhile when an element has more than one common oxidation state. For example Na_2O could be called sodium(I) oxide, but as we know Na always has oxidation state +1, it is perfectly adequate to call it simply sodium oxide.



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Oxidation state and formal charge (introduced in Chapter 4) are both useful tools for electron book-keeping, but they are based on very different premises. Oxidation state is derived by assuming bonds are ionic unless they are between the same element; formal charge is assigned by assuming bonds are pure covalent. Each model has its usefulness and its limitations, as discussed on pages 193 and 415.

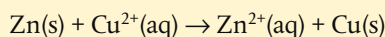
Redox equations

Writing half-equations

Although we have seen that oxidation cannot take place without reduction and vice versa, it is sometimes useful to separate out the two processes from a redox equation and write separate equations for the oxidation and reduction processes. These are therefore called **half-equations**. Electrons are added on one side of each equation to balance the charges.

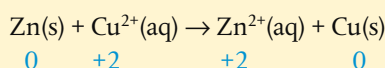
Worked example

Deduce the two half-equations for the following reaction



Solution

Assign oxidation states so we can see what is being oxidized and what is reduced.



Here we can see that Zn is being oxidized and Cu^{2+} is being reduced.

oxidation: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$ **electrons are lost**

reduction: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$ **electrons are gained**

Note there must be equal numbers of electrons in the two half-equations, so that when they are added together the electrons cancel out.

Writing redox equations using half-equations

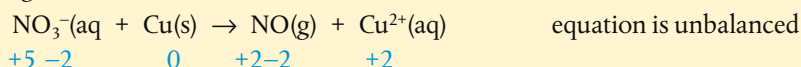
Sometimes we may know the species involved in a redox reaction, but not the overall equation so we need to work this out, making sure it is balanced for both atoms and charge. A good way to do this is to write half-equations for the oxidation and reduction processes separately, and then add these two together to give the overall reaction. Many of these reactions take place in acidified solutions and we therefore use H_2O and/or H^+ ions to balance the half-equations. The process is best broken down into a series of steps, as shown in the example below.

Worked example

Write an equation for the reaction in which NO_3^- and Cu react together in acidic solution to produce NO and Cu^{2+} .

Solution

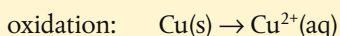
- 1 Assign oxidation states to determine which atoms are being oxidized and which are being reduced:



Therefore Cu is being oxidized ($0 \rightarrow +2$) and N is being reduced ($+5 \rightarrow +2$).

- 2 Write half-equations for oxidation and reduction.

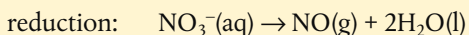
- (a) Balance the atoms other than H and O:



In this example the Cu and N are already balanced.

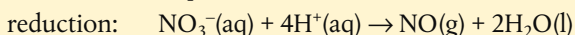
- (b) Balance each half-equation for O by adding H_2O as needed.

Here the reduction equation needs two more O atoms on the right-hand side:

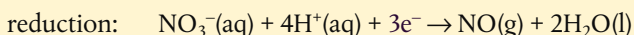
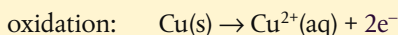


- (c) Balance each half-equation for H by adding H^+ as needed.

Here the reduction equation needs four H atoms on the left-hand side:



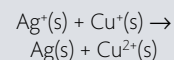
- (d) Balance each half-equation for charge by adding electrons to the side with the more positive charge. (Electrons will be products in the oxidation equation and reactants in the reduction equation.)



- (e) Now check that each half-equation is balanced for atoms and for charge.



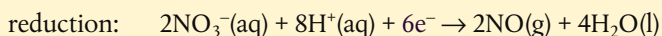
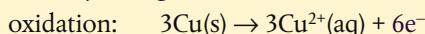
Photochromic lenses darken in response to increasing light intensity and become lighter in dim light. The change is brought about by a redox reaction between silver chloride and copper(I) chloride, which are both embedded in the lens. In bright light Ag^+ ions oxidize Cu^+ forming Ag and Cu^{2+} :



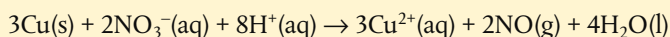
The silver metal produced creates a film over the lens, cutting out glare. In dim light the reaction reverses as Cu^{2+} oxidizes Ag, and the lens clears.

- 3** Equalize the number of electrons in the two half-equations by multiplying each appropriately.

Here the equation of oxidation must be multiplied by 3, and the equation of reduction by 2, to give six electrons in both equations:



- 4** Add the two half-equations together, cancelling out anything that is the same on both sides, which includes the electrons.



The final equation should be balanced for atoms and charge and have no electrons.



Summary of steps in writing redox equations.

- 1** Assign oxidation states to determine which atoms are being oxidized and which are being reduced.
- 2** Write half-equations for oxidation and reduction as follows:
 - (a) balance the atoms other than H and O;
 - (b) balance each half-equation for O by adding H_2O as needed;
 - (c) balance each half-equation for H by adding H^{+} as needed;
 - (d) balance each half-equation for charge by adding electrons to the sides with the more positive charge.
 - (e) check that each half-equation is balanced for atoms and for charge.
- 3** Equalize the number of electrons in the two half-equations by multiplying each appropriately.
- 4** Add the two half-equations together, cancelling out anything that is the same on both sides.

Exercises

- 1** Assign oxidation states to all elements in the following compounds:

(a) NH_4^{+}	(d) SO_2	(g) MnO_2	(i) $\text{K}_2\text{Cr}_2\text{O}_7$
(b) CuCl_2	(e) Fe_2O_3	(h) PO_4^{3-}	(j) MnO_4^{-}
(c) H_2O	(f) NO_3^{-}		
- 2** Use oxidation states to deduce which species is oxidized and which is reduced in the following reactions:
 - (a) $\text{Sn}^{2+}(\text{aq}) + 2\text{Fe}^{3+}(\text{aq}) \rightarrow \text{Sn}^{4+}(\text{aq}) + 2\text{Fe}^{2+}(\text{aq})$
 - (b) $\text{Cl}_2(\text{aq}) + 2\text{NaBr}(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{NaCl}(\text{aq})$
 - (c) $2\text{FeCl}_2(\text{aq}) + \text{Cl}_2(\text{aq}) \rightarrow 2\text{FeCl}_3(\text{aq})$
 - (d) $2\text{H}_2\text{O(l)} + 2\text{F}_2(\text{g}) \rightarrow 4\text{HF(aq)} + \text{O}_2(\text{g})$
 - (e) $\text{I}_2(\text{aq}) + \text{SO}_3^{2-}(\text{aq}) + \text{H}_2\text{O(l)} \rightarrow 2\text{I}^{-}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{H}^{+}(\text{aq})$
- 3** Deduce the half-equations of oxidation and reduction for the following reactions:
 - (a) $\text{Ca(s)} + 2\text{H}^{+}(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
 - (b) $2\text{Fe}^{2+}(\text{aq}) + \text{Cl}_2(\text{aq}) \rightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$
 - (c) $\text{Sn}^{2+}(\text{aq}) + 2\text{Fe}^{3+}(\text{aq}) \rightarrow \text{Sn}^{4+}(\text{aq}) + 2\text{Fe}^{2+}(\text{aq})$
 - (d) $\text{Cl}_2(\text{aq}) + 2\text{Br}^{-}(\text{aq}) \rightarrow 2\text{Cl}^{-}(\text{aq}) + \text{Br}_2(\text{aq})$
- 4** Write balanced equations for the following reactions that occur in acidic solutions:
 - (a) $\text{Zn(s)} + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{SO}_2(\text{g})$
 - (b) $\text{I}^{-}(\text{aq}) + \text{HSO}_4^{-}(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + \text{SO}_2(\text{g})$
 - (c) $\text{NO}_3^{-}(\text{aq}) + \text{Zn(s)} \rightarrow \text{NH}_4^{+}(\text{aq}) + \text{Zn}^{2+}(\text{aq})$
 - (d) $\text{I}_2(\text{aq}) + \text{OCl}^{-}(\text{aq}) \rightarrow \text{IO}_3^{-}(\text{aq}) + \text{Cl}^{-}(\text{aq})$
 - (e) $\text{MnO}_4^{-}(\text{aq}) + \text{H}_2\text{SO}_3(\text{aq}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + \text{Mn}^{2+}(\text{aq})$

5 Which equation represents a redox reaction?

- A** $\text{NaOH(aq)} + \text{HNO}_3\text{(aq)} \rightarrow \text{NaNO}_3\text{(aq)} + \text{H}_2\text{O(l)}$
B $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
C $\text{CuO(s)} + 2\text{HCl(aq)} \rightarrow \text{CuCl}_2\text{(aq)} + \text{H}_2\text{O(l)}$
D $\text{MgCO}_3\text{(s)} + 2\text{HNO}_3\text{(aq)} \rightarrow \text{Mg(NO}_3)_2\text{(aq)} + \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$

6 The oxidation state of chromium is the same in all the following compounds except:

- A** CrCl_3 **B** Cr_2O_3 **C** $\text{Cr}_2(\text{CO}_3)_3$ **D** CrO_3

7 Deduce the names of the following compounds, using their oxidation numbers:

- (a)** Cr_2O_3 **(c)** HNO_3 **(d)** HNO_2 **(e)** PbO_2
(b) CuCl

NATURE OF SCIENCE

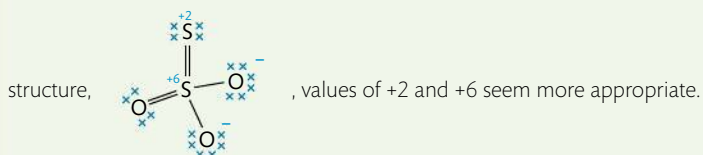
We have seen that oxidation states have several important uses as they enable us to:

- track redox changes in reactions;
- apply systematic names to compounds;
- determine reacting ratios in redox reactions.

At the same time, because they do not have a structural basis, oxidation states can raise some potential misunderstandings or ambiguities.

- Oxidation states do not distinguish between partial and complete transfer of electrons: +4 for C in CO_2 or +7 for Mn in KMnO_4 do not represent losses of 4 and 7 electrons respectively.
- In some cases, especially in organic chemistry, oxidation states have fractional values: in propane, C_3H_8 , the oxidation state of C is $-2\frac{2}{3}$.
- Oxidation states and Lewis (electron dot) structures sometimes suggest conflicting information: CH_4 , C_2H_6 , and C_3H_8 are all similarly bonded molecules, but their C atoms have different oxidation states, -4, -3, and $-2\frac{2}{3}$ respectively.

In $\text{S}_2\text{O}_3^{2-}$, the rules would suggest oxidation states of +2 on each S atom, but from the Lewis



Overall, oxidation states are best considered as a very useful tool that chemists use widely, while also keeping an eye on their limitations.

Oxidizing and reducing agents

We have seen that redox reactions *always* involve the simultaneous oxidation of one reactant with the reduction of another as electrons are transferred between them. The reactant that accepts electrons is called the **oxidizing agent** as it brings about oxidation of the other reactant. In the process it becomes reduced. Likewise the reactant that supplies the electrons is known as the **reducing agent**, because it brings about reduction and itself becomes oxidized. Sometimes the terms **reductant** and **oxidant** are used in place of reducing agent and oxidizing agent respectively.



Molten iron being tapped from a blast furnace for transfer to the steel-making furnace. Iron is extracted from its ore Fe_2O_3 by reducing it with carbon in the form of coke. Although blast furnaces existed in China from about the 5th century BCE and were widespread across Europe, a major development occurred in England in 1709. Substitution of the reducing agent charcoal for coke produced a less brittle form of iron. This accelerated the iron trade and was a key factor in the British Industrial Revolution.

In a redox equation the substance that is reduced is the oxidizing agent, the substance that is oxidized is the reducing agent.

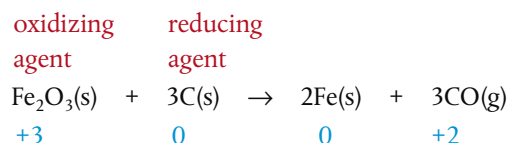
CHALLENGE YOURSELF

1 Hydrogen peroxide, H_2O_2 , can also act as an oxidizing agent and as a reducing agent. With reference to the oxidation states of its atoms, suggest why this is so and which behaviour is more likely.

Single replacement reactions involving metals with metal ions

Full details of how to carry out this experiment with a worksheet are available online.

For example, in the reaction where iron (Fe) is extracted from its ore (Fe_2O_3):



(The oxidation state of O is not shown as it does not change during the reaction.)

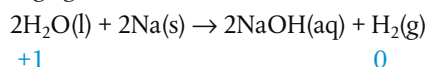
The reducing agent C brings about the reduction of Fe ($+2 \rightarrow 0$), while it is oxidized to CO ($0 \rightarrow +2$). The oxidizing agent Fe_2O_3 brings about the oxidation of C, while itself is reduced to Fe.

Some examples of useful oxidizing and reducing agents are given below:

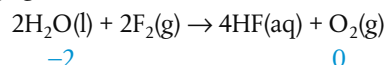
- oxidizing agent: O_2 , O_3 , H^+/MnO_4^- , $H^+/Cr_2O_7^{2-}$, F_2 , Cl_2 , conc. HNO_3 , H_2O_2
- reducing agent: H_2 , C , CO , SO_2 , reactive metals

Note that whether a species acts as an oxidizing or as a reducing agent actually depends on what it is reacting with. For example, water can act as an oxidizing agent and be reduced to hydrogen, for example by sodium, or act as a reducing agent and be oxidized to oxygen, for example by fluorine.

- H_2O acting as an oxidizing agent:



- H_2O acting as a reducing agent:



In fact, water is weak both as an oxidizing agent and as a reducing agent, which is why it is such a useful solvent for many redox reactions.

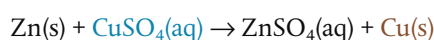


Treatment of drinking water to kill pathogens commonly uses oxidizing agents such as chlorine and ozone. The process is effective and relatively cheap, yet millions of people worldwide lack access to clean drinking water. 'Access to safe water is a fundamental human need and therefore a human right.' Kofi Annan, United Nations Secretary General. In 2010 this human right was passed into international law, but this does not yet make it a reality for millions of people in developing countries.

More reactive metals are stronger reducing agents

Of course not all oxidizing and reducing agents are of equal strength. Some will be stronger than others, depending on their relative tendencies to lose or gain electrons. We learned in Chapter 4 that metals have a tendency to lose electrons and form positive ions, so they will act as reducing agents, pushing their electrons on to another substance. More reactive metals lose their electrons more readily and so we might expect that they will be stronger reducing agents than less reactive metals.

We can check this out by seeing if one metal is able to reduce the ions of another metal in solution. If we immerse zinc in a solution of copper sulfate, a reaction occurs. The blue colour of the solution fades, the pinkish-brown colour of copper metal appears and there is a rise in temperature. What is happening is that the Cu^{2+} ions are being displaced from solution as they are reduced by Zn. At the same time Zn dissolves as it is oxidized to Zn^{2+} .



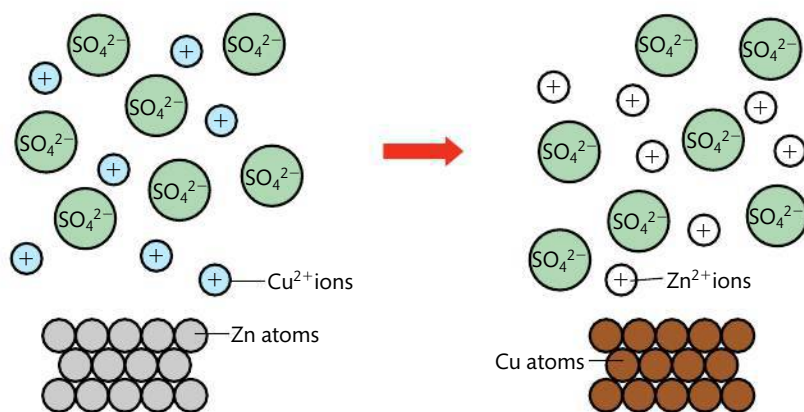
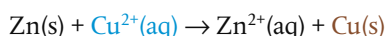


Figure 9.1 The reaction of zinc with copper(II) sulfate solution.

We can write this as an ionic equation without showing the sulfate ions as these act as spectator ions and are not changed during the reaction.



So zinc has acted as the reducing agent – it is therefore the more reactive metal. We can think of it as having the reducing strength to ‘force’ copper ions to accept the electrons.

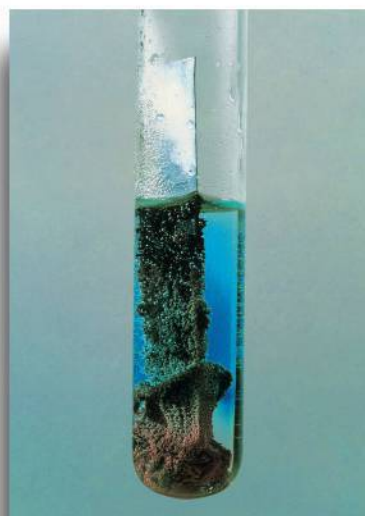
We could confirm this by trying the reaction the other way round, with copper metal immersed in a solution of zinc ions. Can you predict what will happen? The answer is there will be no reaction, because Cu is not a strong enough reducing agent to reduce Zn^{2+} . This is another way of saying that it is a less reactive metal, less able to push the electrons onto Zn^{2+} .

By comparing **displacement** reactions like these between different combinations of metals and their ions, we can build up a list of relative strengths of the metals as reducing agents. This is called the **activity series** and it enables us to predict whether a particular redox reaction between a metal and the ions of another metal will be feasible. Later in this chapter we will learn how to quantify these differences in metal reactivity, which is of great importance in many industrial processes. For example, the extraction of a metal from its ore often involves choosing a suitable reducing agent by reference to these data.

Here is a small part of the activity series of metals:

Mg	strongest reducing agent, most readily becomes oxidized
Al	
Zn	
Fe	
Pb	
Cu	
Ag	weakest reducing agent, least readily becomes oxidized

A simple activity series is given in section 25 of the IB data booklet and is not something that needs to be learned. The important thing is that you are able to interpret it and to deduce it from given data.



A strip of zinc metal half submerged in a solution of CuSO_4 . Solid copper that appears brown is deposited and the blue colour fades as the copper ions are reduced by the zinc. Zinc is shown to be the stronger reducing agent.



More reactive metals are stronger reducing agents than less reactive metals.



A more reactive metal is able to reduce the ions of a less reactive metal.

Redox reactions between a metal and the oxide of a less reactive metal can also confirm the activity series. Here powdered aluminium reduces iron oxide, forming aluminium oxide and iron.

$2\text{Al(s)} + \text{Fe}_2\text{O}_3\text{(s)} \rightarrow \text{Al}_2\text{O}_3\text{(s)} + 2\text{Fe(s)}$. This shows that aluminium is a stronger reducing agent than iron. The reaction is exothermic, creating short bursts of extremely high temperatures.



Worked example

Refer to the activity series given on page 417 to predict whether the following reactions will occur:

- (a) $\text{ZnCl}_2\text{(aq)} + 2\text{Ag(s)} \rightarrow 2\text{AgCl(s)} + \text{Zn(s)}$
 (b) $2\text{FeCl}_3\text{(aq)} + 3\text{Mg(s)} \rightarrow 3\text{MgCl}_2\text{(aq)} + 2\text{Fe(s)}$

Solution

- (a) This reaction would involve Ag reducing Zn^{2+} in ZnCl_2 . But Ag is a weaker reducing agent than Zn, so this will not occur.
 (b) This reaction involves Mg reducing Fe^{3+} in FeCl_3 . Mg is a stronger reducing agent than Fe, so this will occur.

We can investigate how some non-metals such as carbon and hydrogen would fit into this activity series of metals by similar types of displacement reactions. Carbon is able to reduce the oxides of iron and metals below it in the series, which provides one of the most effective means for the extraction of these metals. The position of hydrogen relative to the metals is discussed in section 19.1 on pages 432–435.

More reactive non-metals are stronger oxidizing agents

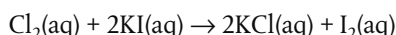
In a similar way, the different strengths of non-metals as oxidizing agents can be compared. For example, the halogens (Group 17 elements) react by gaining electrons and forming negative ions, and so act as oxidizing agents by removing electrons from other substances. We learned in Chapter 3 that their tendency to do this decreases down the group, so we would expect the following trend.

More reactive non-metals are stronger oxidizing agents than less reactive non-metals.

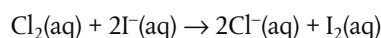


F	strongest oxidizing agent, most readily becomes reduced
Cl	
Br	
I	weakest oxidizing agent, least readily becomes reduced

Again this can be verified by reacting one halogen with solutions containing the ions of another halogen (known as halide ions). For example:



Here the Na^+ ions are spectator ions so we can write the ionic equation without showing them:



The reaction occurs because Cl is a stronger oxidizing agent than I and so is able to remove electrons from it. In simple terms, you can think of it as a competition for electrons where the stronger oxidizing agent, in this case chlorine, will always 'win'.

Exercises

- 8 Identify the oxidizing agents and the reducing agents in the following reactions:
- (a) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$
 - (b) $2\text{Al}(\text{s}) + 3\text{PbCl}_2(\text{s}) \rightarrow 2\text{AlCl}_3(\text{s}) + 3\text{Pb}(\text{s})$
 - (c) $\text{Cl}_2(\text{aq}) + 2\text{KI}(\text{aq}) \rightarrow 2\text{KCl}(\text{aq}) + \text{I}_2(\text{aq})$
 - (d) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
- 9 Use the two reactivity series given to predict whether reactions will occur between the following reactants, and write equations where relevant.
- (a) $\text{CuCl}_2 + \text{Ag}$
 - (b) $\text{Fe}(\text{NO}_3)_2 + \text{Al}$
 - (c) $\text{NaI} + \text{Br}_2$
 - (d) $\text{KCl} + \text{I}_2$
- 10 (a) Use the following reactions to deduce the order of reactivity of the elements W, X, Y, Z, putting the most reactive first.
- $$\text{W} + \text{X}^+ \rightarrow \text{W}^+ + \text{X}$$
- $$\text{X} + \text{Z}^+ \rightarrow \text{X}^+ + \text{Z}$$
- $$\text{Y}^+ + \text{Z} \rightarrow \text{no reaction}$$
- $$\text{X} + \text{Y}^+ \rightarrow \text{X}^+ + \text{Y}$$
- (b) Which of the following reactions would you expect to occur according to the reactivity series you established in part (a)?
- (i) $\text{W}^+ + \text{Y} \rightarrow \text{W} + \text{Y}^+$
 - (ii) $\text{W}^+ + \text{Z} \rightarrow \text{W} + \text{Z}^+$



Chlorine gas bubbling through a clear solution of potassium iodide KI. The solution is turning brown due to the formation of iodine in solution, as chlorine oxidizes the iodide ions and forms chloride ions.



A more reactive non-metal is able to oxidize the ions of a less reactive non-metal.

Redox titrations

As with acid–base titrations (discussed in Chapter 8), redox titrations are used to determine the unknown concentration of a substance in solution. But in this case, the technique is based on a redox reaction between the two reactants, and finds the equivalence point where they have reacted stoichiometrically by transferring electrons. The two types of titration are compared below.

Acid–base titration	Redox titration
neutralization reaction between acid and base	redox reaction between oxidizing agent and reducing agent
protons are transferred from acid to base	electrons are transferred from reducing agent to oxidizing agent

Redox titrations are carried out in much the same way as acid–base titrations, using a burette and pipette to measure volumes accurately, and a standard solution of

CHALLENGE YOURSELF

- 2 Reactions in which the same element is simultaneously oxidized and reduced is known as **disproportionation**. Show how this happens in the reaction when Cl_2 and NaOH react to produce NaCl , NaClO , and H_2O .



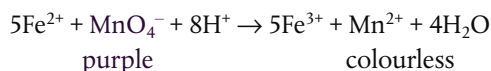
The browning of fruit when it is exposed to the air is an oxidation reaction. It occurs when phenols released in damaged cells are oxidized with molecular oxygen at alkaline pH. The brown products are tannins. Food chemists are often interested in slowing down these reactions using reducing agents such as SO_2 or vitamin C, for example in lemon juice.

one reactant. An indicator is usually used to signal the equivalence point, although some redox changes are accompanied by a colour change and may not need an external indicator. From the volume of the solution added from the burette to reach equivalence, which is known as the **titre**, the concentration of the other reactant can be determined. The calculations are based on the redox equation, which is typically developed from the half-equations.

Redox titrations are commonly used in the food and beverages industry, in the pharmaceutical industry, and in water and environmental analysis. For example, wines can be analysed for the presence of sulfur dioxide, and the vitamin C content of foods can be determined in this way. Some examples of common redox titrations that can be carried out in the laboratory are given below, and more details on how to carry out these experiments can be found online.

1 Analysis of iron with manganate(VII)

This redox titration uses KMnO_4 in an acidic solution as the oxidizing agent, which oxidizes Fe^{2+} ions to Fe^{3+} . During the reaction MnO_4^- is reduced to Mn^{2+} , so the overall equation is:



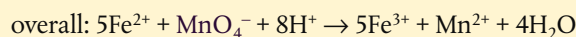
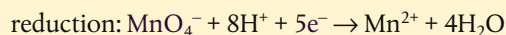
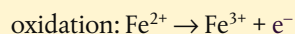
The reaction is accompanied by a colour change from deep purple to colourless, so the reaction acts as its own indicator, signalling the equivalence point.

Worked example

All the iron in a 2.000 g tablet was dissolved in an acidic solution and converted to Fe^{2+} . This was then titrated with KMnO_4 . The titration required 27.50 cm^3 of $0.100 \text{ mol dm}^{-3} \text{ KMnO}_4$. Calculate the total mass of iron in the tablet and its percentage by mass. Describe what would be observed during the reaction, and how the equivalence point can be detected.

Solution

First we need the balanced equation for the reaction, which is solved by the half-equation method described earlier.

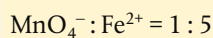


Next we need to know the amounts of reactants used to reach equivalence. We start with KMnO_4 as we know both its concentration and its volume.

$$n = cV$$

$$n(\text{MnO}_4^-) = 0.100 \text{ mol dm}^{-3} \times \frac{27.50}{1000} \text{ dm}^3 = 0.00275 \text{ mol MnO}_4^-$$

From the equation for the reaction we know the reacting ratio:



$$\therefore \text{mol Fe}^{2+} = 0.00275 \text{ mol MnO}_4^- \times 5 = 0.01375 \text{ mol}$$

Redox titration

Full details of how to carry out this experiment with a worksheet are available online.



$$M(\text{Fe}) = 55.85 \text{ g mol}^{-1}$$

$$n = \frac{m}{M}$$

$$\therefore m(\text{Fe}) = 0.01375 \text{ mol} \times 55.85 \text{ g mol}^{-1} = 0.7679 \text{ g}$$

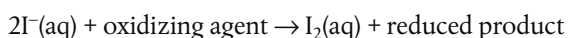
$$\% \text{ Fe in tablet} = \frac{0.768 \text{ g}}{2.000 \text{ g}} \times 100 = 38.39\%$$

$$\therefore \text{Fe in tablet} = 0.768 \text{ g}, 38.4\%$$

MnO_4^- in the burette is purple, but forms a nearly colourless solution in the flask as it reacts to form Mn^{2+} . But when the reducing agent Fe^{2+} in the flask has been used up at equivalence, MnO_4^- ions will not react and the purple colour will persist.

2 Iodine–thiosulfate reaction

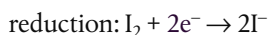
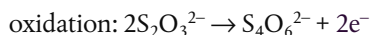
Several different redox titrations use an oxidizing agent to react with excess iodide ions to form iodine.



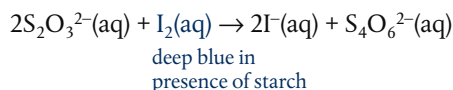
Examples of oxidizing agents used in this way include KMnO_4 , KIO_3 , $\text{K}_2\text{Cr}_2\text{O}_7$, and NaOCl .

The liberated iodine, I_2 , is then titrated with sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, using starch as an indicator.

Redox equations:



The overall equation is:



The starch indicator is added during the titration (not at the start) and forms a deep blue colour by forming a complex with free I_2 . As the I_2 is reduced to I^- during the reaction, the blue colour disappears, marking the equivalence point.

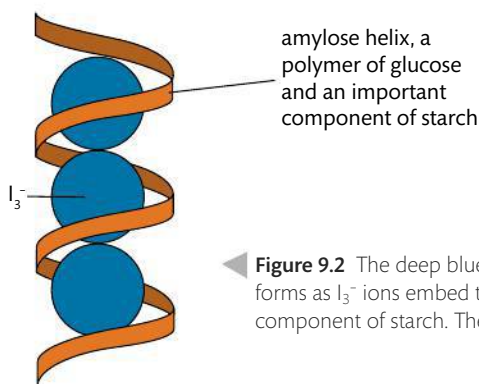


Figure 9.2 The deep blue complex between iodine solution and starch forms as I_3^- ions embed themselves within the helix of amylose, a component of starch. The structure of I_3^- was discussed in Chapter 4.



Because iodine, I_2 , has a very low solubility in water, it is used instead in the form of 'iodine solution', which is made by dissolving I_2 in iodide ions, forming the complex ion $\text{I}_3^-(\text{aq})$. It is the I_3^- ion that gives the deep blue complex with starch. Even though I_3^- is the reactant, it is still valid to use I_2 in the equations, as this is the 'active ingredient'. Technically, starch is not a redox indicator as such, as it responds to the disappearance of I_2 , rather than to a change in the redox potential of the solution.

CHALLENGE YOURSELF

- 3** From your knowledge of the structure of I_3^- , consider why it forms a stable complex in the hydrophobic interior of the amylose helix.

In titration calculations, always look first for the reactant where you know both its reacting volume and its concentration. Start by using these data to calculate the number of moles. Then use the reacting ratio in the equation to deduce the moles of the other reactant.

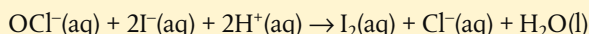


A canal in England covered in a bloom of duckweed. This is an aquatic weed that thrives in water that contains pollutants such as phosphates. This is known as eutrophication. The bloom blocks out sunlight to deeper layers of the water ecosystem, and so lowers the level of dissolved oxygen.



Worked example

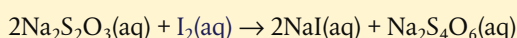
Household bleach is an oxidizing agent that contains sodium hypochlorite, NaOCl, as the active ingredient. It reacts with iodide ions in acidic solution as follows:



A 10.00 cm³ sample of bleach was reacted with a solution of excess iodide ions, and the liberated iodine was then titrated with Na₂S₂O₃. The titration required 38.65 cm³ of 0.0200 mol dm⁻³ Na₂S₂O₃. Determine the concentration of OCl⁻ in the bleach.

Solution

First we need the balanced equation for the titration, which is solved by the half-equation method on page 413.

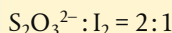


Next we calculate the number of moles of Na₂S₂O₃ as we know both its concentration and its volume.

$$n = cV$$

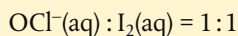
$$n(\text{S}_2\text{O}_3^{2-}) = 0.0200 \text{ mol dm}^{-3} \times \frac{38.65}{1000} \text{ dm}^3 = 7.73 \times 10^{-4} \text{ mol S}_2\text{O}_3^{2-}$$

From the equation for the reaction we know the reacting ratio



$$\therefore \text{mol I}_2 = 7.73 \times 10^{-4} \times 0.5 = 3.865 \times 10^{-4}$$

This is a back-titration as the I₂ was liberated by the reaction of I⁻ with OCl⁻, as given in the question. The reacting ratio is



$$\therefore \text{mol OCl}^- = \text{mol I}_2 = 3.865 \times 10^{-4}$$

The concentration of the OCl⁻ in the bleach can now be calculated.

$$n(\text{OCl}^-) = cV$$

$$\therefore 3.865 \times 10^{-4} \text{ mol} = c \times \frac{10.00}{1000} \text{ dm}^3$$

$$[\text{OCl}^-] = 0.0387 \text{ mol dm}^{-3}$$

3 Winkler method for calculating dissolved oxygen

The **dissolved oxygen** content of water is one of the most important indicators of its quality, as oxygen is essential to the survival of aquatic life. As the level of pollution in water increases, the dissolved oxygen content generally decreases, as the oxygen is used by bacteria in decomposition reactions. The **biological oxygen demand** (BOD) is therefore used as a means of measuring the degree of pollution. BOD is defined as the amount of oxygen used to decompose the organic matter in a sample of water over a specified